



Week 4: Descriptive statistics/Exploratory Data Analysis

Geog4300/6300: Harper

Sept 9, 2025

Where are we?

- ◆ Reading catch-up:
 - ◆ Wickham and Grolemand Ch 1-3, 5, 7 (optional: 4, 6, 8)
 - ◆ Wikham and Grolemand Ch 21 (Databases)
- ◆ This week: Read McGrew et al. Chapter 3 (Descriptive statistics and graphs)
- ◆ Work on scripts on your own time if we're not getting to them in class. Ask questions during 'lab' time (5:10-5:35 both days) or book a time with me or Ummie.
- ◆ Lab 1: Due Thursday

Know your variable types

- **Nominal/categorical:** non-ordered and qualitative
 - Geographic regions, land use classifications
- **Ordinal:** qualitative but ranked
 - Months of the year, storm intensity
- **Interval:** Quantitative but without a meaningful zero
 - Temperature, years
- **Ratio:** Quantitative and divisible
 - Rainfall amounts, program budget amounts

What kind of data is it?

- Letter grades on an essay exam
- Birth year of graduate students in a department
- Number of students from each country in a large course
- Degree objectives for geography students (BA/BS, MA/MS, Phd)
- Flavors of ice cream
- Grams of fat in ice cream
- % of consumers who prefer particular flavors of ice cream

- **Nominal/categorical:** non-ordered and qualitative
- **Ordinal:** qualitative but ranked
- **Interval:** Quantitative but without a true zero
- **Ratio:** Quantitative and divisible

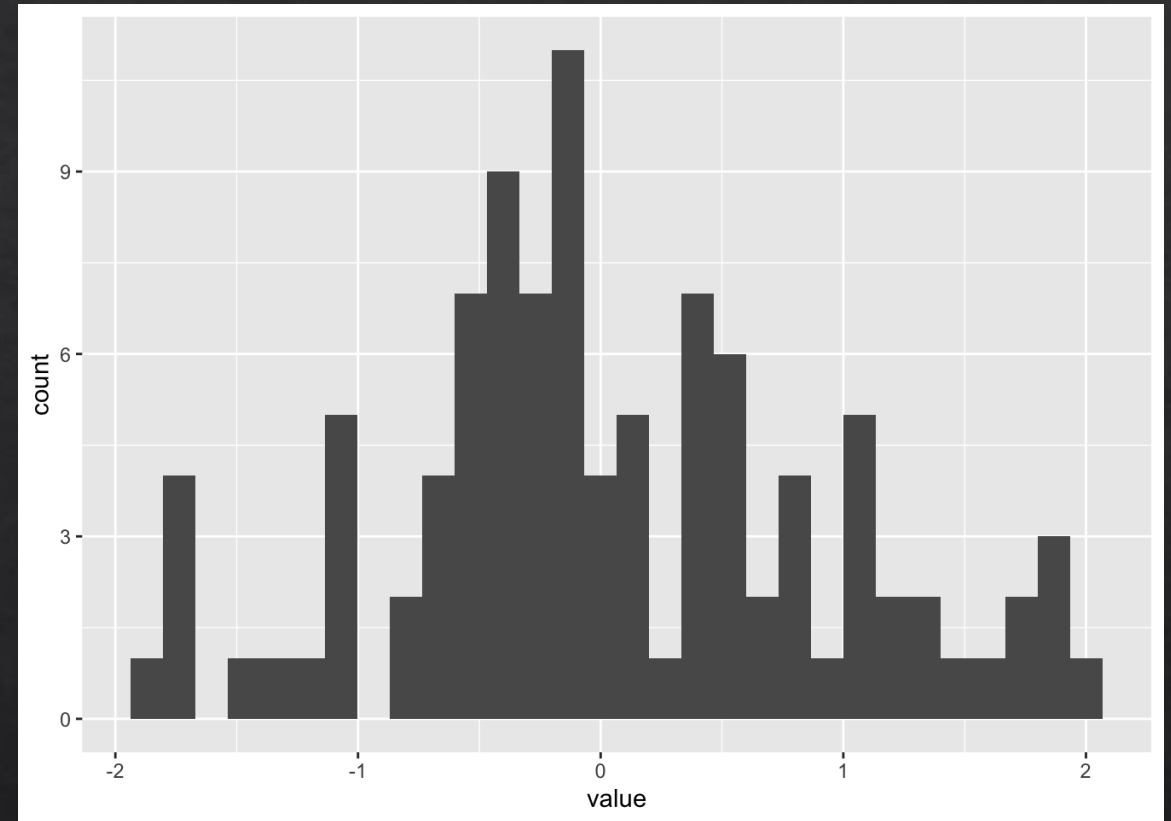
Descriptive statistics

Measures of central tendency:

- Mean
- Median
- Mode

Measures of distribution:

- Standard deviation
- Interquartile range (IQR)
- Coefficient of variation



Which measure of central tendency should we use for the number of UGA football games we have attended? (And how should we wrangle this data before doing so?)

0	0
4	0
3	0
0.5 (I left at halftime)	0
Seven home games	0
1	0
13	0
2	4
Almost all home games since 2005, plus 3 or 4 bowl games	3
0	

What's a related question we could ask for which mode would be the appropriate measure of central tendency?



Coefficient of variation

Standard deviation divided by the mean

Useful when means vary across groups or measures

$$CV = \frac{s}{\bar{x}}$$

R challenge!

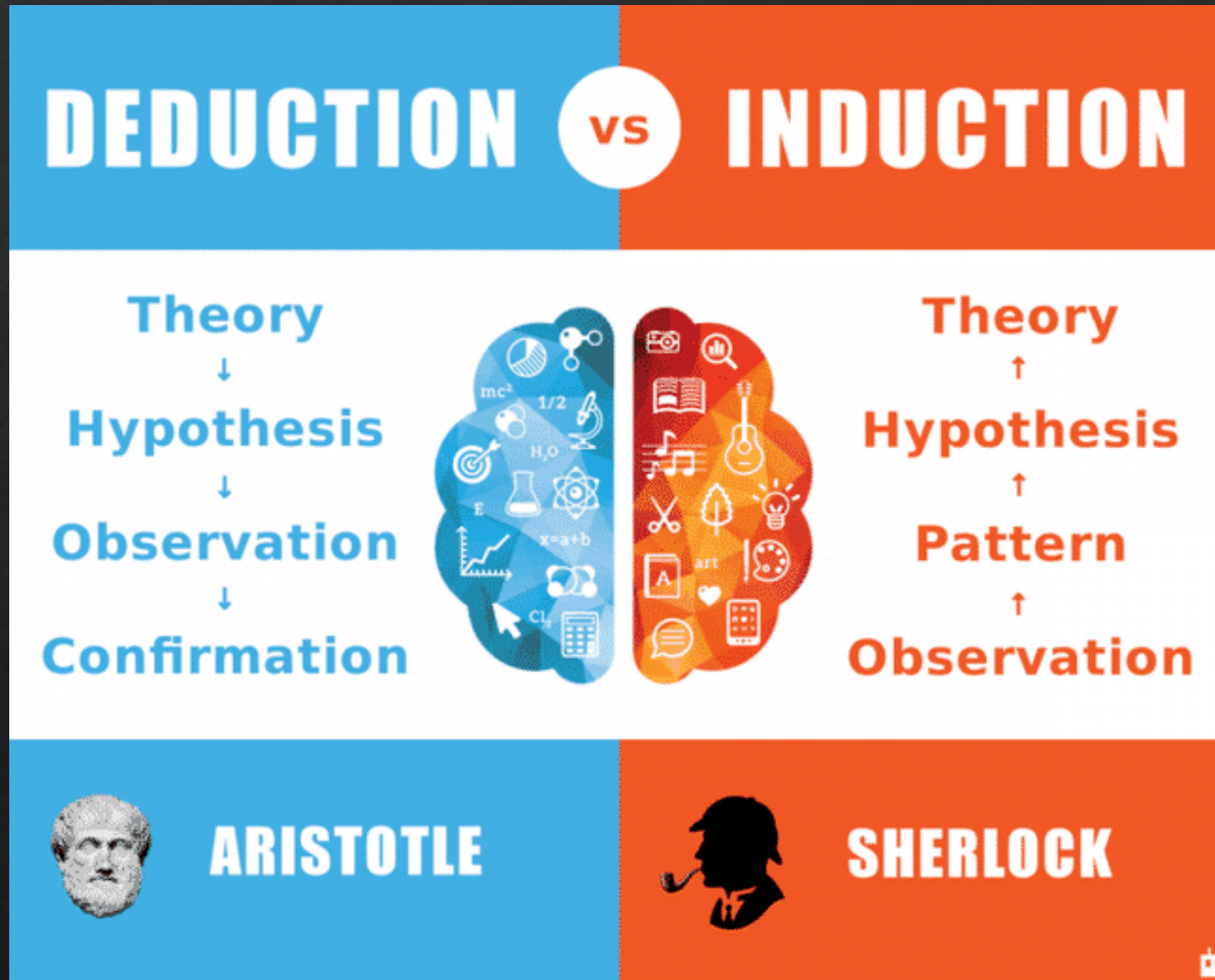
Load the Daymet data in the class repo:
`daymet_monthly_median_2010-2021`

Calculate the mean, median, standard deviation (sd), and cv for prcp by county. You'll need to use `group_by`, `summarise`, and `mutate`. Which counties have the highest cv? How does that compare to standard deviation?

$$CV = \frac{s}{\bar{x}}$$



What is exploratory data analysis?



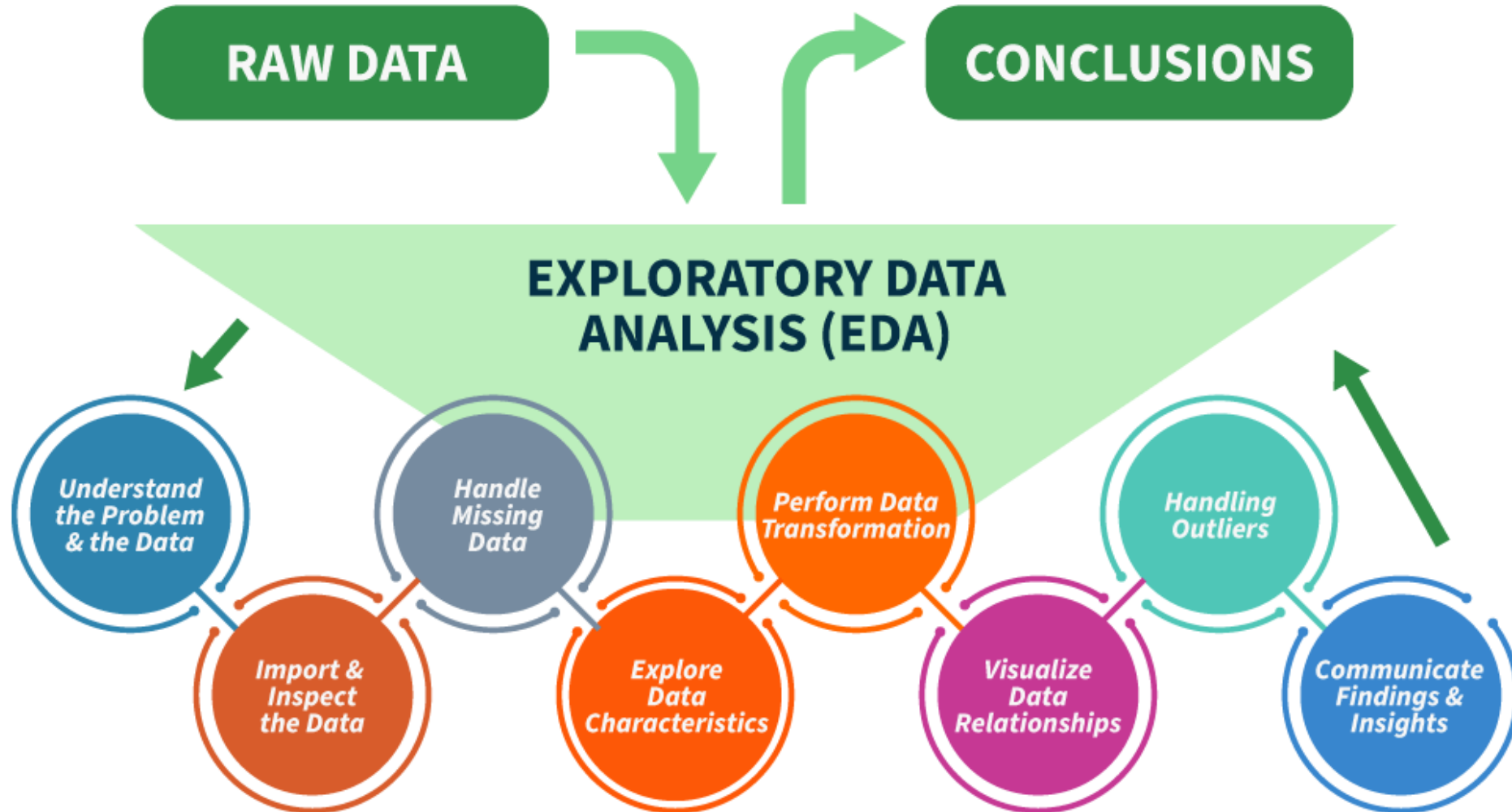
What is exploratory data analysis?



John Tukey

“Numerical quantities focus on expected values, graphical summaries on unexpected values.”

Steps for Performing Exploratory Data Analysis



Tukey and Wickham on Exploratory Data Analysis

